UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II **Advanced Calculus (2AC)**

Functions of several variables: Differentiability

Dr Francois Ebobisse Bille

0.1 Partial derivatives

Definition 0.1 Let $f : \mathbb{R}^2 \to \mathbb{R}$ and let $(a, b) \in \mathbb{R}^2$.

(i) We say that f has partial derivative w.r.t. x at the point (a, b) if the limit $\lim\limits_{x\to 0} \dfrac{f(a + x, b) - f(a, b)}{x}$ exists (as a real number) and

$$\boxed{\frac{\partial f}{\partial x}(a, b) := \lim_{x\to 0} \frac{f(a + x, b) - f(a, b)}{x}}$$

is called the partial derivative of f w.r.t. x at the point (a, b).

This means that, in order to check whether the function f has partial derivative w.r.t. x at the point (a, b), we study the differentiability of the function $x \to f(x, b)$ at the number $x = a$ and the derivative of this new function of the only varaiable x at the number $x = a$ (if it does exist) is the partial derivative of the function f w.r.t. x at the point (a, b), denoted by $\dfrac{\partial f}{\partial x}(a, b)$.

Other notations: $f_x(a, b)$ or $D_x f(a, b)$.

(ii) We say that f has partial derivative w.r.t. y at the point (a, b) if the limit $\lim\limits_{y\to 0} \dfrac{f(a, b + y) - f(a, b)}{y}$ exists and

$$\boxed{\frac{\partial f}{\partial y}(a, b) := \lim_{y\to 0} \frac{f(a, b + y) - f(a, b)}{y}}$$

is called the partial derivative of f w.r.t. y at the point (a, b).

This means that, in order to check whether the function f has partial derivative w.r.t. y at the point (a, b), we study the differentiability of the function $y \to f(a, y)$ at the number b, denoted by $\dfrac{\partial f}{\partial y}(a, b)$..

Other notations: $f_y(a, b)$ or $D_y f(a, b)$.

Example 0.2 (Partial derivatives versus Continuity)

(a) Consider the function $f(x,y) = \sqrt{x^2+y^2}$. We want to study whether f has both partial derivatives at the point $(0,0)$.

For the partial derivative w.r.t. x we want to check whether the limit $\lim_{x\to 0} \frac{f(x,0)-f(0,0)}{x}$ exists. We find that

$$\boxed{\lim_{x\to 0} \frac{f(x,0)-f(0,0)}{x} = \lim_{x\to 0} \frac{|x|}{x} = \begin{cases} 1 & \text{when } x \to 0^+, \\ -1 & \text{when } x \to 0^-. \end{cases} \quad \Rightarrow \quad f_x(0,0) \text{ does not exist.}}$$

By symmetry, also $f_y(0,0)$ does not exist. Notice that the function f is continuous at $(0,0)$.

(b) On the other hand, consider the function

$$\boxed{f(x,y) := \begin{cases} \dfrac{xy}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}}$$

Notice that

$$\boxed{f_x(0,0) = \lim_{x\to 0} \frac{f(x,0)-f(0,0)}{x} = \lim_{x\to 0} \frac{0-0}{x} = 0 \quad \text{and similarly} \quad f_y(0,0) = 0}.$$

However, the limits along each axis is equal to 0 while the limit along the line $y = x$ is equal to $\frac{1}{2}$ since $f(x,x) = \frac{1}{2}$. Therefore, the f is not continuous at $(0,0)$ (in fact the limit $\lim_{(x,y)\to(0,0)} f(x,y)$ does not even exist) while both partial derivatives $f_x(0,0) = f_y(0,0) = 0$.

The first example illustrates the fact that a function can be continuous at a point whithout having partial derivatives at that point. On the other hand, the second example illustrates the fact that a function can have partial derivative at a point while not being continuous at that point (in fact notice that the function does not even have limit).

0.2 Geometric interpretation of partial derivatives

Let $f : \mathbb{R}^2 \to \mathbb{R}$ and let (a,b) be such that $f_x(a,b)$ and $f_y(a,b)$ exist.

Consider the curve $\mathcal{C}_1$ intersection of the graph $z = f(x,y)$ with the vertical plane $y = b$. The curve $\mathcal{C}_1$ is described by the vector function $\vec{r}_1(x) := (x, b, f(x,b))$ and is regular around the point $(a, b, f(a,b))$. Its tangent line at that point has direction $\vec{r}_1'(a) = (1, 0, f_x(a,b))$.

On the other hand, let $\mathcal{C}_2$ be the curve intersection of the graph $z = f(x,y)$ with the vertical plane $x = a$. The curve $\mathcal{C}_2$ is decribed by the vector function $\vec{r}_2(y) := (a, y, f(a,y))$, is regular around the point $(a, b, f(a,b)$ and its tangent line at that point has direction $\vec{r}_2'(b) = (0, 1, f_y(a,b))$. Notice that

$$\boxed{\vec{r}_1'(a) \times \vec{r}_2'(b) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f_x(a,b) \\ 0 & 1 & f_y(a,b) \end{vmatrix} = (-f_x(a,b), -f_y(a,b), 1) \neq (0,0,0)}.$$

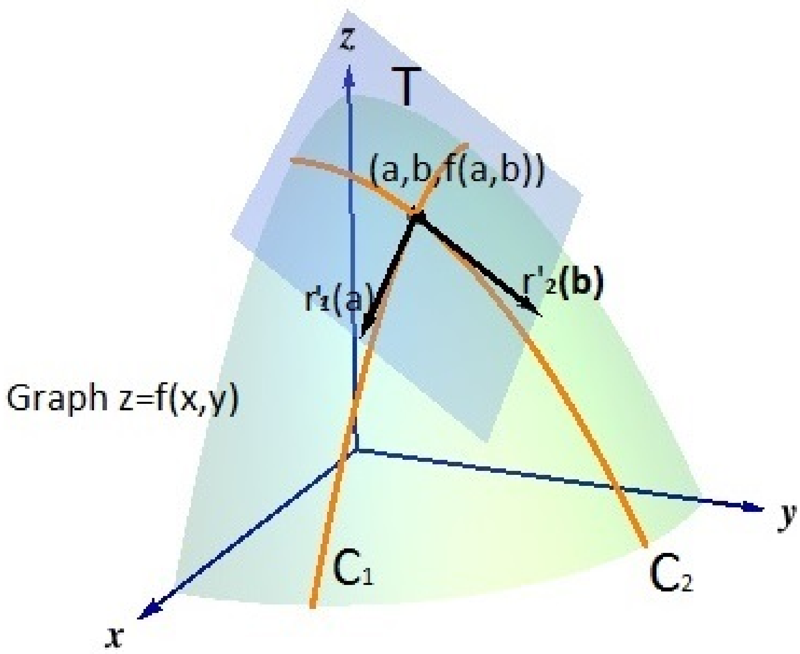

Figure 1: The plane T determined by the vectors $\vec{r}_1'(a) = (1, 0, f_x(a, b))$ and $\vec{r}_2'(b) = (0, 1, f_y(a, b))$. The picture extracted from web.ma.utexas.edu/users/m408m/Display14-4-2.shtml, has been conveniently modified following our notations.

Therefore, the two vectors $\vec{r}_1'(a)$ and $\vec{r}_2'(b)$ determine a plane T passing through the point $(a, b, f(a, b))$ and whose normal vector is given by $\vec{r}_1'(a) \times \vec{r}_2'(b)$ (see Figure 1). So, the plane T has cartesian equation

$$\begin{aligned}
& (-f_x(a,b), -f_y(a,b), 1) \cdot (x-a, y-b, z-f(a,b)) = 0 \\
& -f_x(a,b) \times (x-a) - f_y(a,b) \times (y-b) + 1 \times (z - f(a,b)) = 0 \\
\Leftrightarrow \quad & \boxed{z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)}\,.
\end{aligned}$$

We are now in a position to define the notion of differentiability of a function of two variables at a given point.

0.3 Differentiability - Tangent plane

Definition 0.3 (Differentiability)
Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function for which both partial derivatives $f_x(a, b)$ and $f_y(a, b)$ exist. The

function f is said to be differentiable at the point (a, b) if

$$\boxed{\lim_{(x,y)\to(a,b)} \frac{f(x,y) - f(a,b) - f_x(a,b)(x-a) - f_y(a,b)(y-b)}{\sqrt{(x-a)^2 + (y-b)^2}} = 0} \tag{1}$$

Remark 0.4

(a) Notice that the numerator of the left hand-side in (1) is the difference between "the graph" of f and the "plane T" while the denominator is the distance between the running point (x, y) and the fixed point (a, b).

(b) Notice that if the function f is differentiable at the point (a, b) i.e., (1) holds, then the numerator in (1) has limit equal to 0, since the denominator has limit equal to 0. So we get that

$$\begin{aligned}
&\lim_{(x,y)\to(a,b)} \big[f(x,y) - f(a,b) - f_x(a,b)(x-a) - f_y(a,b)(y-b)\big] = 0 \\
\Rightarrow\ &\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b) + \lim_{(x,y)\to(a,b)} \big[f_x(a,b)(x-a) + f_y(a,b)(y-b)\big] \\
=\ &f(a,b) + 0 = f(a,b)\,.
\end{aligned}$$

Hence, $\boxed{\text{the function } f \text{ is continuous at the point } (a,b)}$.

(c) Also, from the fact that $\lim_{(x,y)\to(a,b)} \big[f(x,y) - f(a,b) - f_x(a,b)(x-a) - f_y(a,b)(y-b)\big] = 0$, it follows that

$$f(x,y) - f(a,b) - f_x(a,b)(x-a) - f_y(a,b)(y-b) \approx 0 \quad \text{for } (x,y) \text{ close enough to the point } (a,b)\,.$$

That is,

$$\boxed{\begin{array}{c} f(x,y) \approx f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) \\ \text{for } (x,y) \text{ close enough to the point } (a,b) \end{array}} \qquad \textbf{(LINAPP)}_1$$

is called **the linear approximation of the function f around** (a, b). Using the gradient $\nabla f(a, b)$, we can also write $\textbf{(LINAPP)}_1$ in the form

$$\boxed{\begin{array}{c} f(x,y) \approx f(a,b) + \nabla f(a,b) \cdot (x-a, y-b) \\ \text{for } (x,y) \text{ close enough to the point } (a,b) \end{array}} \qquad \textbf{(LINAPP)}_2$$

Example 0.5 Let us study the differentiability at the point $(0, 0)$ of the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$\boxed{f(x,y) := \begin{cases} \dfrac{x^2 y}{x^2 + y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0)\,. \end{cases}}$$

From the definition of differentiability in (1) we will first see whether the function f has both partial derivatives at the point $(0, 0)$.
For the partial derivative w.r.t. x, we need to find the limit (if it does exist)

$$\boxed{\lim_{x\to 0} \frac{f(x,0) - f(0,0)}{x} = \lim_{x\to 0} \frac{0-0}{x} = 0 = f_x(0,0) \quad \text{and similarly } f_y(0,0) = 0}\,.$$

Now according to (1), it remains to check whether (or not)

$$\boxed{\lim_{(x,y)\to(0,0)} \frac{f(x,y) - f(0,0) - f_x(0,0)(x-0) - f_y(0,0)(y-0)}{\sqrt{(x-0)^2 + (y-0)^2}} = 0} \,. \tag{2}$$

Notice that

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y) - f(0,0) - f_x(0,0)(x-0) - f_y(0,0)(y-0)}{\sqrt{(x-0)^2 + (y-0)^2}} = \lim_{(x,y)\to(0,0)} \frac{x^2 y}{(x^2+y^2)^{3/2}}$$

$$= \begin{cases} 0 & \text{along the axes} \\ \displaystyle\lim_{x\to 0^+} \frac{x^3}{2\sqrt{2}x^3} = \frac{1}{2\sqrt{2}} \neq 0 & \text{along } y = x > 0 \end{cases}$$

Hence, the function f is not differentiable because the limit in (2) does exist.

Exercises 0.6

1. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x,y) := \begin{cases} \dfrac{xy^3}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0)\,. \end{cases}$$

   (a) Study the continuity of the function f at the point $(0,0)$.
   Hint: You could write $|f(x,y)| = |xy| \times \dfrac{y^2}{x^2+y^2}$.

   (b) Find (if they exist) the partial derivatives $f_x(0,0)$ and $f_y(0,0)$.

   (c) Say whether the function f is differentiable at $(0,0)$.

2. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x,y) := \begin{cases} \dfrac{xy^2 - y^3}{x^4+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0)\,. \end{cases}$$

   (a) Study the continuity of the function f at the point $(0,0)$.
   Hint: You could write $|f(x,y)| = |x-y| \times \dfrac{y^2}{x^4+y^2}$.

   (b) Find (if they exist) the partial derivatives $f_x(0,0)$ and $f_y(0,0)$.

   (c) Is the function f differentiable at $(0,0)$? Explain clearly your answer.

3. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x,y) := \begin{cases} (x^2+y^2)\sin\left(\dfrac{1}{x^2+y^2}\right) & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0)\,. \end{cases}$$

(a) Study the continuity of the function f at the point $(0,0)$.
Hint: $|\sin(u)| \leq 1$.

(b) Find (if they exist) the partial derivatives $f_x(0,0)$ and $f_y(0,0)$.

(c) Say whether the function f is differentiable at $(0,0)$.

4. Consider the function $f(x,y) := \arctan\left(\sqrt{4x^2+y^2}\right)$.

   (a) Find the gradient $\nabla f(0,1)$ of f at the point $(0,1)$.
   Hint: Find $f_x(0,1)$ and $f_y(0,1)$ by differentiating.

   (b) Write down the cartesian equation of the tangent plane to the graph of the function f at the point $(0,1)$.

   (c) Write down the linear approximation of the function f around the point $(0,1)$.

   (d) Deduce an approximation of $f(-0.02, 0.94)$.

5. Consider the function $f(x,y) := x\left(\sqrt{x^2+y^2}\right)$.

   (a) Study the differentiability of f at every point $(x,y) \in \mathbb{R}^2$.

   (b) Show that both partial derivatives f_x and f_y are continuous at every $(x,y) \in \mathbb{R}^2$.

Definition 0.7 If $f : \mathbb{R}^2 \to \mathbb{R}$ has partial derivatives at (a,b), then the vector

$$\boxed{\nabla f(a,b) = (f_x(a,b), f_y(a,b))}$$

is called the *gradient* of f at the point (a,b) .
In general, for $f : \mathbb{R}^n \to \mathbb{R}$, then

$$\boxed{\nabla f(x) = (f_{x_1}(x), f_{x_2}(x), \dots, f_{x_n}(x))}$$

is the gradient of f at the point $x = (x_1, x_2, \dots, x_n)$

Remark 0.8 The differentiability of the function $f : \mathbb{R}^n \to \mathbb{R}$ at the point $a = (a_1, a_2, \dots, a_n)$ is formulated in terms of gradient as

$$\boxed{\lim_{x \to a} \frac{f(x) - f(a) - \nabla f(a) \cdot (x-a)}{\|x-a\|} = 0}$$

where $\quad \|x-a\| := \sqrt{(x_1-a_1)^2 + \ldots + (x_n-a_n)^2}$.

0.3.1 The Chain Rule

We recall that the Chain Rule **(CR)** is about the differentiability of a composition of differentibale functions.

Theorem 0.9 (The Chain Rule)
Let $\vec{r} : (\alpha, \beta) \to \mathbb{R}^n$ be differentiable at $t_0 \in (\alpha, \beta)$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\vec{r}(t_0)$. Then the function $f \circ \vec{r}$ is also differentiable at t_0 with

$$\boxed{\frac{d}{dt}(f \circ \vec{r})(t_0) = \nabla f(\vec{r}(t_0)) \cdot \vec{r}\,'(t_0)}\,. \tag{3}$$

Proof: $\vec{r} : (\alpha, \beta) \to \mathbb{R}^n$ is differentiable at t_0, then we can write the linear approximation of $\vec{r}$ about the point t_0 to get

$$\boxed{\vec{r}(t) \approx \vec{r}(t_0) + \vec{r}\,'(t_0)(t - t_0) \quad \text{for } t \text{ close enough to } t_0}\,. \tag{4}$$

On the other hand, being f differentiable at $\vec{r}(t_0)$, we have that

$$\boxed{f(\vec{x}) \approx f(\vec{r}(t_0) + \nabla f(\vec{r}(t_0)) \cdot (\vec{x} - \vec{r}(t_0)) \quad \text{for } \vec{x} \text{ close enough to } \vec{r}(t_0)}\,. \tag{5}$$

Now, we can take $\vec{x} = \vec{r}(t)$ in (5) for t close enough to t_0 (**Why is this possible?**) and get

$$\boxed{f(\vec{r}(t)) \approx f(\vec{r}(t_0) + \nabla f(\vec{r}(t_0)) \cdot (\vec{r}(t) - \vec{r}(t_0)) \quad \text{for } t \text{ close enough to } t_0}\,. \tag{6}$$

Now substitute (4) into (6) and get

$$\boxed{f(\vec{r}(t)) \approx f(\vec{r}(t_0) + \nabla f(\vec{r}(t_0)) \cdot \vec{r}\,'(t_0)(t - t_0) \quad \text{for } t \text{ close enough to } t_0}\,. \tag{7}$$

We want to show that $\displaystyle\lim_{t \to t_0} \frac{(f \circ \vec{r})(t) - (f \circ \vec{r})(t_0)}{t - t_0} = \nabla f(\vec{r}(t_0)) \cdot \vec{r}\,'(t_0)$, which easily follows from (7). In fact

$$\lim_{t \to t_0} \frac{(f \circ \vec{r})(t) - (f \circ \vec{r})(t_0)}{t - t_0} = \lim_{t \to t_0} \frac{f(\vec{r}(t)) - f(\vec{r}(t_0))}{t - t_0} = \nabla f(\vec{r}(t_0)) \cdot \vec{r}\,'(t_0)\,.$$

Have you answered the question after (5)?

■

Remark 0.10 Similary, if $f : \mathbb{R}^2 \to \mathbb{R}$ is differentiable at (a, b) and $g : \mathbb{R} \to \mathbb{R}$ is differentiable at $f(a, b)$, then the composition $g \circ f$ is Differentiable at the point (a, b) with

$$\boxed{\nabla(g \circ f)(a, b) = g'(f(a, b))\nabla f(a, b) \quad \Leftrightarrow \quad \begin{cases} (g \circ f)_x(a, b) = g'(f(a, b))f_x(a, b), \\ \\ (g \circ f)_y(a, b) = g'(f(a, b))f_y(a, b) \end{cases}}$$

.

In fact, we use linear approximation $\textbf{(LINAPP)}_2$ to write

$$f(x, y) \approx f(a, b) + \nabla f(a, b) \cdot (x - a, y - b) \text{ for } (x, y) \text{ sufficiently close to } (a, b). \tag{8}$$

On the other hand, we write

$$g(t) \approx g(f(a, b)) + g'(f(a, b))(t - f(a, b)) \quad \text{for } t \text{ close enough to } f(a, b). \tag{9}$$

Now, taking $t = f(x, y)$ for (x, y) sufficiently close to (a,b) (**Once again, why such a choice can be made here?**), we obtain that

$$\begin{aligned} g(f(x, y)) &\approx g(f(a, b)) + g'(f(a, b))(f(x, y) - f(a, b)) \quad (x, y) \text{ close enough to } (a, b) \\ &\approx g(f(a, b)) + g'(f(a, b))\nabla f(a, b) \cdot (x - a, y - b) \quad (x, y) \text{ close enough to } (a, b)\,. \end{aligned}$$

So,

$$\frac{g(f(x,y)-g(f(a,b))-g'(f(a,b))\nabla f(a,b)\cdot(x-a,y-b)}{\sqrt{(x-a)^2+(y-b)^2}}\approx 0 \qquad (x,y) \text{ close enough to } (a,b).$$

Therefore,

$$\boxed{\lim_{(x,y)\to(a,b)}\frac{g(f(x,y)-g(f(a,b))-g'(f(a,b))\nabla f(a,b)\cdot(x-a,y-b)}{\sqrt{(x-a)^2+(y-b)^2}}=0}$$

which shows that the function $g\circ f$ is differentiable at the point (a,b) with

$$\nabla(g\circ f)(a,b)=g'(f(a,b))\nabla f(a,b)\,.$$

■

Example 0.11 Let $f:\mathbb{R}\to\mathbb{R}$, $\vec{r}:[-2,10]\to\mathbb{R}^2$ and $g:\mathbb{R}\to\mathbb{R}$ be such that

$$\vec{r}(5)=(0,1);\quad \vec{r}'(5)=(3-2)\quad \nabla f(0,1)=-2;\quad \nabla f(0,1)=(\pi,-1);\quad g'(-2)=2\pi\,.$$

Find $(f\circ\vec{r})'(5)$ and $\nabla(g\circ f)(0,1)$.

0.3.2 Directional derivatives

Definition 0.12 (Directional derivative)
Let $f:\mathbb{R}^2\to\mathbb{R}$ and let $\vec{u}\in\mathbb{R}^2$ be a unit vector. We call *directional derivative* of the function f at the point (a,b), the number

$$\boxed{f_{\vec{u}}(a,b):=\lim_{t\to 0}\frac{f(a,b)+t\vec{u})-f(a,b)}{t}\qquad\text{provided the limit exists}\,.}\tag{10}$$

Other notations: $f_{\vec{u}}(a,b)=\partial_{\vec{u}}f(a,b)=D_{\vec{u}}f(a,b)$.

Remark 0.13 According to the definition of the directional derivative above, we only consider the restriction of the function f to the line passing through the point (a,b) with direction the vector $\vec{u}$. That line is described by the vector function $\vec{r}(t):=(a,b)+t\vec{u}$. The directional derivative is then

$$\boxed{f_{\vec{u}}(a,b)=\frac{d}{dt}(f\circ\vec{r})(0)}\,.\tag{11}$$

The identity (11) will be used later in order to establish a more practical way to evaluate the directional derivative $f_{\vec{u}}(a,b)$.

Exercises 0.14

1. Consider the function $f:\mathbb{R}^2\to\mathbb{R}$ defined by $f(x,y):=\begin{cases}\dfrac{x^3y^2}{x^6+y^2} & \text{if } (x,y)\neq(0,0),\\[2ex] 0 & \text{if } (x,y)=(0,0)\,.\end{cases}$

   (a) Study the continuity of the function f at the point $(0,0)$.

(b) Find (if it exists) the directional derivative $f_{\vec{u}}(0,0)$ of f at $(0,0)$ along every direction $\vec{u} = (u_1, u_2) \in \mathbb{R}^2$ with $\|\vec{u}\| = 1$.

(c) Is the function f differentiable at the point $(0,0)$? Explain clearly your answer.

2. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) := \begin{cases} 1 & \text{if } y = x^2 \neq 0, \\ 0 & \text{otherwise.}. \end{cases}$

   (a) Study the continuity of the function f at the point $(0,0)$.

   (b) Find (if it exists) the directional derivative $f_{\vec{u}}(0,0)$ of f at $(0,0)$ along every direction $\vec{u} = (u_1, u_2) \in \mathbb{R}^2$ with $\|\vec{u}\| = 1$. (Hint: Explain why $f(t\vec{u}) = 0$.)

   (c) Is the function f differentiable at the point $(0,0)$? Explain clearly your answer.

3. Consider the function $f(x,y) := \sqrt{y \sin(x^2 + y^2)}$.

   (a) Find (if they exist) the partial derivatives $f_x(0,0)$ and $f_y(0,0)$.

   (b) Find the directional derivative $f_{\vec{u}}(0,0)$ at every direction $\vec{u}$.

   (c) Study the differentiability of f at the point $(0,0)$.

Proposition 0.15 *Let $f : \mathbb{R}^2 \to \mathbb{R}$ be differentiable at the point (a,b). Then,*

$$\boxed{f_{\vec{u}}(a,b) = \nabla f(a,b) \cdot \vec{u}}. \tag{12}$$

Proof: It follows from (11) and the Chain Rule (noticing that $\vec{r}(0) = (a,b)$ and $\vec{r}'(0) = \vec{u}$) that

$$\boxed{f_{\vec{u}}(a,b) = \frac{d}{dt}(f \circ \vec{r})(0) \underbrace{=}_{(CR)} \nabla f(\vec{r}(0)) \cdot \vec{r}'(0) = \nabla f(a,b) \cdot \vec{u}}.$$

■

Example 0.16 Consider the function $f(x,y) := \arctan(e^x + x^2 + 3y)$. Find the directional derivative $f_{\vec{u}}(0,0)$ of the function f at the point $(0,0)$, where $\vec{u} = (-1/\sqrt{2}, 1/\sqrt{2})$.
We will use the identity (11) for which we need

$$\boxed{f_x(x,y) = \frac{e^x + 2x}{1 + (e^x + x^2 + 3y)^2}, \quad f_y(x,y) = \frac{3}{1 + (e^x + x^2 + 3y)^2} \quad \Rightarrow \quad f_x(0,0) = \frac{1}{2}, \quad f_y(0,0) = \frac{3}{2}}.$$

So,

$$f_{\vec{u}}(0,0) = \nabla f(0,0) \cdot \vec{u} = \left(\frac{1}{2}, \frac{3}{2}\right) \cdot \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = -\frac{1}{2\sqrt{2}} + \frac{3}{2\sqrt{2}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

Remark 0.17 Notice that the choice of functions of two variables is not restrictive. All the properties presented above can be extended to functions of more than two variables.

Exercise 0.18 Consider the function $f(x,y,z) = e^{xyz} \sin(xz)$. Find the directional derivative $f_{\vec{u}}(1, \pi, 2)$ in the direction of $\vec{u} = (3, 1, 4)$.
Hint: Remember to rather use the unit vector in the direction of the given vector.

Theorem 0.19 (Directions of max/min increase)
Let the function $f : \mathbb{R}^2 \to \mathbb{R}$ be differentiable at the point (a, b) such that $\nabla f(a, b) \neq (0, 0)$. Then,

(a) the vector $\boxed{\vec{u} := \dfrac{\nabla f(a,b)}{\|\nabla f(a,b)\|}}$ *indicates the direction of maximum increase of the function f at the point (a, b) and the corresponding increase is equal to* $\boxed{\|\nabla f(a,b)\|}$.

(b) the vector $\boxed{\vec{v} := -\dfrac{\nabla f(a,b)}{\|\nabla f(a,b)\|}}$ *indicates the direction of maximum decrease of the function f at the point (a, b) and the corresponding decrease is equal to* $\boxed{-\|\nabla f(a,b)\|}$.

Proof: We want to find $\vec{u} \in \mathbb{R}^2$ unit vector such that the directional derivative $f_{\vec{u}}(a, b)$ is maximum/minimum. Notice that

$$f_{\vec{u}}(a,b) = \nabla f(a,b) \cdot \vec{u} = \|\nabla f(a,b)\|\|\vec{u}\| \cos\theta = \|\nabla f(a,b)\| \cos\theta \,.$$

Now, $\cos\theta$ is maximum when $\theta = 0$, which means that the vector $\vec{u}$ is the unit vector in the direction of $\nabla f(a, b)$, that is, $\vec{u} = \dfrac{\nabla f(a,b)}{\|\nabla f(a,b)\|}$. In that case, the maximum value of $\cos\theta = 1$. Therefore, the maximum value of $f_{\vec{u}}$ is $\|\nabla f(a, b)\|$.

On the other hand $\cos\theta$ is minimum for $\theta = \pi$ and the minimum value is -1 which corresponds to the fact that the vector $\vec{u}$ is the unique vector in the direction opposite to $\nabla f(a, b)$, that is $\vec{u} = -\dfrac{\nabla f(a,b)}{\|\nabla f(a,b)\|}$ for which $f_{\vec{u}}(a, b) = -\|\nabla f(a, b)\|$.

■

Example 0.20 Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x,y) := \begin{cases} \dfrac{x - xy}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0)\,, \\[2ex] 0 & \text{if } (x,y) = (0,0)\,. \end{cases}$$

(a) Study the continuity of the function f at the point $(0, 0)$.
Hint: You could use polar coordinates r, θ through $x = r\cos\theta,\ y = r\sin\theta$..

(b) Say whether the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ exist. Find them if they exist.

(c) Say whether the function f is differentiable at the point $(0, 0)$. Explain clearly your answer.

Solution:

(a) We want to check whether $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0) = 0$. Along the x-axis, we get

$$\lim_{x\to 0^+} f(x,0) = \lim_{x\to 0^+} \frac{x}{x} = 1 \text{ while } \lim_{x\to 0^-} f(x,0) = \lim_{x\to 0^-} \frac{x}{-x} = -1\,.$$

So, the function f is not even continuous at $(0, 0)$ along the x-axis.

(b) For the partial derivative w.r.t. x, we need find (if it does exist) the limit

$$\lim_{x\to 0}\frac{f(x,0)-f(0,0)}{x}=\lim_{x\to 0}\frac{x}{|x|x}=\lim_{x\to 0}\frac{1}{|x|}=\infty\,.$$

Hence, $f_x(0,0)$ does not exist.

For the partial derivative w.r.t. y, we need find (if it does exist) the limit

$$\lim_{y\to 0}\frac{f(0,y)-f(0,0)}{y}=\lim_{x\to 0}\frac{0}{|y|y}=0\,.$$

Hence, $f_y(0,0)=0$.

(c) The function f is not differentaible at $(0,0)$ since the partial derivative $f_x(0,0)$ does not exist.

Exercises 0.21

1. Consider the function $f:\mathbb{R}^2\to\mathbb{R}$ defined by

$$f(x,y):=\begin{cases}\dfrac{|\sin x|y^2}{x\sqrt{x^2+y^2}} & \text{if } x\neq 0,\\[2ex] 0 & \text{if } x=0\,.\end{cases}$$

   (a) Study the continuity of the function f at every point on the x-axis. (**Hint:** $|\sin x|\leq |x|$ and $|y|\leq\sqrt{x^2+y^2}$).

   (b) Say whether the partial derivatives $f_x(0,0)$ and $f_y(0,0)$ exist. Find them if they exist. (**Answer:** $f_x(0,0)=f_y(0,0)=0$).

   (c) Say whether the function f is differentiable at the point $(0,0)$. Explain clearly your answer.

2. Write down the cartesian equation of the tangent plane to the graph of the function $f(x,y)=\arctan(xy)$ and the point $(1,1,\frac{\pi}{4})$.
(**Hint:** $z=f(1,1)+f_x(1,1)(x-1)+f_y(1,1)(y-1)$).

The next set of notes will be on Taylor formula for functions of several variables!